GRADUATION PROJECT DOCUMENTATION

SMARTLISTING COPILOT

Unified Literature Review

Computer Vision, Natural Language Processing & Multimodal Listing Consistency

An Evidence-Based Review of Fashion Product Attribute Recognition and Explainable Image–Text Consistency Checking

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Evidence note. All numerical performance findings in this chapter are results reported in prior publications under their original datasets and evaluation protocols. SmartListing Copilot has not yet produced experimental results.

Abstract

Fashion e-commerce listings can be inconsistent when product images and seller-provided text imply different product types or colors. This integrative literature review synthesizes six Computer Vision studies and five studies covering short-text attribute extraction, multimodal product analysis, and explainability. It examines fashion recognition, transfer learning, ResNet, MobileNet, CLIP zero-shot classification, garment-color estimation, dictionary-assisted NLP, gated multimodal fusion, attribute generation, and interpretable prediction. Across these areas, published methods primarily evaluate recognition or attribute completion rather than an explicit decision about whether image-derived evidence contradicts a listing's textual claims.

The proposed SmartListing Copilot workflow addresses that distinction by retaining separate visual predictions and text-extracted claims before an explainable comparison stage. Within the approved project scope, Fashion Product Images Small supplies images and the styles.csv fields id, articleType, baseColour, and productDisplayName; exactly five product categories will be chosen after exploratory data analysis. Manually authored or synthetic descriptions compensate for the absence of a description column. Proposed visual candidates are transfer-learned ResNet or MobileNet models, with CLIP zero-shot as a parallel baseline; the language branch is rule-based, English-first, and restricted to a small Arabic lexicon. Contradictions and omissions lead to reviewer-facing reason codes and template-based suggestions requiring seller approval.

The central research gap is the absence, among the selected studies, of validated, transparent Product Type and Color contradiction detection under these specific data and workflow constraints. Evidence is compared within its original tasks and datasets, without pooling incompatible metrics. The chapter identifies evaluation requirements for visual reliability, textual claim validity, mismatch precision, calibration, and manual review. No performance results are claimed for SmartListing Copilot.

Keywords — Fashion e-commerce; Computer Vision; Natural Language Processing; product attribute extraction; multimodal consistency; contradiction detection; explainable decision support; transfer learning.

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1. Introduction and Integrated Research Objectives

Fashion product listings contain at least two distinct forms of evidence: a product image depicting the item and language that claims what the item is. A visually plausible image does not establish that the seller's title is accurate; similarly, a well-written title does not establish that its color and product type match the image. A listing-quality system must therefore estimate what the image supports, recover what the language asserts, and explicitly compare the two rather than treating modality agreement as a given.

SmartListing Copilot is proposed as a bounded decision-support workflow for this setting. Its sole MVP data source is Fashion Product Images Small, joined through id to styles.csv. Product Type (articleType) and Product Color (baseColour) are the only core semantic attributes; productDisplayName supplies titles, while descriptions will be manually authored or synthetically constructed because the dataset has no dedicated description column. Exactly five product categories will be confirmed only after exploratory data analysis (EDA). These choices are scope constraints, not findings of the reviewed literature.

The visual branch is expected to compare ResNet and MobileNet under transfer learning and retain CLIP zero-shot classification as a parallel baseline and possible fallback. The language branch uses an English-first controlled vocabulary and rules, with only a small fixed Arabic lexicon. Outputs remain independent until an explicit discrepancy comparator applies deterministic rules and, subject to validation, a lightweight classifier. Warnings include evidence, confidence and reasons, with template-based suggestions subject to seller Accept, Edit or Ignore. Nothing is published automatically; no large language model is required.

The scientific question uniting the two research contributions is whether reliable, explainable contradiction detection can be achieved under restrictive data, attribute and human-review conditions when the available literature predominantly optimizes either visual recognition, extraction from noisy text, or fusion for attribute completion. This chapter examines what the studies actually establish, where their evaluations differ, and which proposed capabilities remain to be tested.

1.1 Review questions

- What do fashion recognition, transfer learning and calibration studies establish about visual Product Type and Color evidence?
- Which methods and documented failure modes matter when extracting Product Type and Color claims from short, noisy listing text?
- Why do existing multimodal attribute-completion results not directly establish image–text contradiction-detection performance?
- Which evidence supports a transparent reviewer-facing explanation and seller-controlled correction flow?
- What experimental checks are needed to distinguish genuine performance from dataset artifacts and synthetic-evaluation leakage?

2. Review Methodology, Evidence Boundaries and Scope

This is a purposive integrative literature review, not a systematic review or meta-analysis. Eleven principal discussions were retained: five foundational Computer Vision publications plus the supplementary garment-color study, and five studies spanning product-attribute NLP, multimodal attribute extraction and explainability. The Computer Vision material also cites calibration methodology and the project dataset. The original reviews used publication records, full manuscripts where accessible, citation tracing and topic-led searches; no claim of exhaustive database coverage or PRISMA screening is made.

Source verification depth is not uniform. The original Computer Vision review checked its transfer-learning and calibration sources at abstract level, whereas the remaining key CV discussions used full-text sources. The MobileNet source is an arXiv preprint and CoRe is a short CVPR workshop paper. The Rakuten study is presented as an arXiv preprint rather than an independently verified journal/conference article. Peer-reviewed work, workshop work, preprints, and data-source records are distinguished, not assigned equal evidential status.

Reported performance is retained only under the original paper's task, split, dataset and metric. Category top-k accuracy, token-label accuracy, accuracy of value prediction, Macro-F1, recall at fixed precision, color-distance threshold rates, and simulated human-trust scores answer different questions. Cross-paper comparisons in this chapter are therefore analytical rather than numerical rankings. Source-specific statements are distinguished from the proposed SmartListing Copilot methodology, which has not been experimentally validated.

The approved dataset is approximately 44,000 products according to the proposal, but the precise number of usable, correctly linked images, category frequencies and image dimensions must be verified during EDA. A secondary report of 80 × 60-pixel images appears in the CV review; this is a provisional description rather than a verified measurement of the available files. The evaluation cannot assume real marketplace descriptions or real-world moderation outcomes.

3. Computer Vision: Fashion Representation, Transfer and Visual Evidence

Recognition of Product Type and Color is an upstream evidence-generation task, not a verdict on listing truth. The following six studies span domain-specific fashion labels, transfer across datasets, convolutional backbone design, visual–language pretraining and the special difficulty of color. Their contributions motivate candidate models and anticipated risks; none measures SmartListing Copilot's downstream contradiction detection.

3.1 Liu et al. (2016): DeepFashion and Rich Clothing Supervision [1]

Research objective. The authors argue that existing clothes datasets carry limited annotations and cannot cope with real-world variation in style, deformation, occlusion and shooting scenario. They build a large dataset with unified annotations and show, with a new model, that such annotations improve clothes recognition and retrieval. The problem matters because clothes classification, attribute prediction and item retrieval serve online shopping directly.

Proposed methodology. Images were collected from two shopping websites (1,320,078 images of 391,482 items) and from Google Images using 12,654 item-name queries (1,273,150 images). Near and exact

duplicates were found by comparing AlexNet fc7 responses, and human annotators then removed unusable low-resolution or off-topic images, leaving 800,000 images. The proposed model, FashionNet, follows VGG-16 up to the penultimate convolutional layer, then splits into a global branch, a local branch that pools features at estimated clothing landmarks, and a pose branch that predicts landmark location and visibility. It is trained iteratively with four losses (landmark regression, softmax for visibility and category, weighted cross-entropy for attributes, triplet loss for pairs). Training pretrains on a subset of 300,000 images (50,000 for validation) and then fine-tunes on each benchmark; the authors state that no fashion item overlaps between fine-tuning and testing sets.

Dataset used. DeepFashion contains over 800,000 images with 50 fine-grained categories, 1,000 attributes in five groups (texture, fabric, shape, part, style), 4 to 8 landmarks per image and more than 300,000 cross-pose or cross-domain pairs. Category labels were assigned manually (one per image). Attribute labels were not annotated manually: an attribute counts as present if it matches the image's metadata text, followed by a human screening pass to remove falsely matched images. About 0.5% of manually labelled samples were inconsistent between two annotators and required a third. The category and attribute benchmark uses 63,720 images. The five attribute groups do not include a dedicated color group.

Models and techniques. FashionNet is compared with two earlier deep models, WTBI (an MLP on top of pretrained ImageNet features) and DARN (an attribute-regularized two-stream network), both trained with clothing bounding boxes and the same data and protocol as FashionNet. Ablations train FashionNet with 100 or 500 attributes instead of 1,000, and replace clothing landmarks with human joints or poselets.

Evaluation metrics and results. Category classification is scored by top-k accuracy and attribute prediction by top-k recall. Reported (Table 2 of [1]): top-3 / top-5 category accuracy is 43.73 / 66.26 for WTBI, 59.48 / 79.58 for DARN and 82.58 / 90.17 for FashionNet; top-3 / top-5 attribute recall over all groups is 27.46 / 35.37, 42.35 / 51.95 and 45.52 / 54.61 respectively. Training with 100 or 500 attributes lowers FashionNet's category top-5 accuracy to 70.57 and 77.39. Replacing landmarks with joints or poselets costs 6 to 9 points. The authors also report that retrieval accuracy for "shorts" was the lowest and for "dress" the highest of the categories analyzed, which they attribute to shorts having similar shapes and plain textures. Note: top-k scores are ranking metrics over 50 categories and are not comparable to single-label top-1 accuracy or to the F1-based metrics planned for SmartListing Copilot.

Strengths. The dataset was, to the authors' knowledge, the largest and most richly annotated clothes dataset at the time. The benchmarks define clear protocols, the item-level split avoids leakage between fine-tuning and test sets, and the ablations show that auxiliary supervision (many attributes, clothing-specific landmarks) improves fine-grained category recognition.

Limitations. Attribute labels are derived from metadata text and only screened, so they are weak labels. FashionNet depends on landmark annotations and a VGG-16-scale network. Cross-domain retrieval stays hard: FashionNet's top-20 consumer-to-shop accuracy is 0.188. The authors identify style variation, deformation, occlusion and scenario change as open difficulties. In the portions reviewed, results are top-k only; per-class precision, recall or F1 are not reported.

Relevance to SmartListing Copilot. (i) The "shorts" observation is concrete evidence that low inter-class distinctiveness hurts clothing recognition, supporting the proposal's decision to limit the system to five categories. (ii) The label-provenance discussion (manual categories versus metadata-derived attributes) is a reminder to audit articleType and baseColour rather than trust them, as the proposal's noisy-color-label risk already anticipates. (iii) The item-level split precedent supports the proposal's split by Product ID before any augmentation or synthetic error injection. (iv) FashionNet's landmark pooling is not applicable, because the proposal lists no landmark fields for the dataset. (v) Because DeepFashion has no color attribute group, it provides no direct evidence for color recognition. (vi) The gain from joint attribute supervision is a hypothesis for the proposed multi-head design (type and color heads on one backbone), but whether a color head helps the type head on this dataset is an untested question.

Synthesis. Rich garment landmarks and large clothing ontologies explain what specialized fashion representations can learn, but the approved small dataset has no landmark annotations and supports only type and color labels. DeepFashion's top-k category scores should not be used as targets for five-class Macro-F1.

3.2 Kornblith et al. (2019): Transferability of ImageNet Features [2]

Research objective. Transfer learning is described as a cornerstone of computer vision, yet little work had examined the relationship between architecture and transfer. The authors test the implicit hypothesis that models that perform better on ImageNet necessarily perform better on other vision tasks.

Proposed methodology. ImageNet-pretrained networks are evaluated in two transfer regimes: as fixed feature extractors and fine-tuned. The study relates ImageNet accuracy to transfer accuracy, examines how the way a network is trained on ImageNet (for example the regularization used) affects transfer, and checks whether pretraining helps on small fine-grained datasets. Further methodological detail was not reviewed.

Dataset used. The abstract refers to image classification datasets, including two small fine-grained ones. The later CLIP study (reviewed in Section 3.5) describes a "12 dataset evaluation suite" from this work. None of the datasets is a fashion catalogue dataset.

Models and techniques. A range of ImageNet classification architectures, used with frozen features (a linear classifier on penultimate-layer features) or fine-tuned end to end.

Evaluation metrics and results. Reported: the correlation between ImageNet accuracy and transfer accuracy is $r = 0.99$ when networks are fixed feature extractors and $r = 0.96$ when they are fine-tuned. In the fixed-feature setting the relationship is very sensitive to ImageNet training: many common forms of regularization slightly improve ImageNet accuracy but yield features that are much worse for transfer. On two small fine-grained datasets, ImageNet pretraining provided minimal benefit. The authors conclude that architectures generalize well across datasets, but ImageNet features are less general than previously suggested.

Strengths. The study is systematic across architectures and across both common transfer regimes, and it turns a widely held assumption into a measured claim. Its practical message, that the checkpoint and the training recipe matter for frozen features, is directly actionable.

Limitations. Datasets are natural-image benchmarks rather than fashion catalogue images. Correlation across a set of architectures is not a guarantee for any single task. Behavior at very low resolution is not addressed in the abstract. Because only the abstract was verified here, details such as which datasets benefit least are not reported in this document.

Relevance to SmartListing Copilot. (i) The pairing of strong ImageNet-to-fine-tuned correlation with weaker, recipe-dependent frozen-feature transfer supports treating fine-tuning, not only frozen features, as a main candidate; a frozen-backbone variant remains useful as a cheap reference. (ii) Because the pretraining recipe affects frozen features, the exact pretrained checkpoint used should be recorded in the experiment log (the proposal plans basic MLflow logging). (iii) The minimal pretraining benefit on small fine-grained datasets is a caution that fashion types and colors at low resolution may gain less from ImageNet pretraining than expected. This is a hypothesis to test, for instance by comparing pretrained against randomly initialized training of the same backbone on the project's training split.

Synthesis. The correlations reported across network families are evidence for comparing checkpoints, not evidence that any pretrained backbone will succeed at the project's effective resolution. Fine-tuning must be compared against frozen features using the same held-out product IDs.

3.3 He et al. (2016): Residual Learning and ResNet [3]

Research objective. Deeper networks should represent more, yet the authors observe a degradation problem: as depth grows, accuracy saturates and then falls, and the added layers raise training error, so the cause is optimization rather than overfitting. The problem is important because depth is central to visual recognition quality.

Proposed methodology. Stacked layers learn a residual function $F(x) = H(x) - x$, and the block outputs $F(x) + x$ through a shortcut. Identity shortcuts add no parameters or computation; projection shortcuts (1×1 convolutions) are used when dimensions change. For 50, 101 and 152 layers a bottleneck block (1×1, 3×3, 1×1) reduces cost. Training is from scratch on ImageNet with scale augmentation (shorter side sampled from 256 to 480), random 224×224 crops with horizontal flips, per-pixel mean subtraction, the standard color augmentation, batch normalization after each convolution, SGD with batch size 256, weight decay 0.0001, momentum 0.9 and no dropout. For detection and localization, ImageNet-pretrained ResNets are fine-tuned, which is a transfer-learning use of the architecture.

Dataset used. ImageNet 2012 (1,000 classes; 1.28 million training and 50,000 validation images; 100,000 test images scored by a server), CIFAR-10 (50,000 training and 10,000 test images, 10 classes, 32×32 pixels), and PASCAL VOC and COCO for detection. All are natural-object datasets; no fashion data are involved.

Models and techniques. 18-layer and 34-layer plain networks and ResNets, plus ResNet-50, -101 and -152, compared with VGG-16, GoogLeNet, PReLU-net and BN-Inception. Reported cost: 1.8, 3.6, 3.8, 7.6 and 11.3 billion FLOPs (multiply-adds) for ResNet-18, 34, 50, 101 and 152, versus 15.3 and 19.6 billion for VGG-16 and VGG-19.

Evaluation metrics and results. Metrics are top-1 and top-5 error for ImageNet, test error for CIFAR-10 and mAP for detection. Selected ImageNet metrics are tabulated in the technical evidence tables below. The paper further reports a single-model ResNet-152 top-5 validation error of 4.49% and an ensemble top-5 test error of 3.57%. On CIFAR-10, ResNet-110 reaches 6.43% error with 1.7 million parameters, while the 1202-layer network (19.4 million parameters) reaches 7.93%, which the authors attribute to overfitting. Replacing VGG-16 with ResNet-101 in Faster R-CNN gives a 28% relative improvement on COCO (mAP@[.5, .95] from 21.2 to 27.2).

Table 1. Selected ImageNet validation metrics from He et al. [3]; 10-crop testing, not fashion-listing scores.

Model	Top-1 error (%)	Top-5 error (%)	GFLOPs
VGG-16 (authors' test)	28.07	9.33	15.3
Plain-34	28.54	10.02	3.6
ResNet-34	25.03	7.76	3.6
ResNet-50	22.85	6.71	3.8
ResNet-101	21.75	6.05	7.6
ResNet-152	21.43	5.71	11.3

Strengths. The degradation problem is resolved at no extra parameter cost, a family of depths lets practitioners choose a compute budget, and the learned representations generalize to detection and localization. The plain-versus-residual comparison is well controlled, since both have the same depth, width and parameters.

Limitations. The authors leave the cause of plain-network optimization difficulty to future work, and the 1202-layer result shows that excess capacity overfits a small dataset. All experiments use natural-object

images at standard sizes; calibration and fashion-specific behavior are not studied. Compute is higher than for mobile-oriented networks.

Relevance to SmartListing Copilot. (i) The ResNet family provides several depth and compute options for the main model; the CIFAR-10 overfitting result suggests that, for a five-category, low-resolution dataset, the smaller members of the family are the natural starting point (a hypothesis, not a finding for this dataset). (ii) Table 1 of [3] shows a spatial reduction from 224 × 224 to 7 × 7 (a factor of 32); applied to an 80 × 60 input this would leave a feature map of about 3 × 2, so input size is a genuine design variable (native versus resized) rather than a detail. (iii) The training recipe includes color augmentation. Because color is one of the two predicted attributes, hue and saturation jitter could corrupt the color labels and should be excluded for the color head. This is a proposed precaution derived from the recipe, not a result in the paper.

3.4 Howard et al. (2017): MobileNet and Compute-Constrained Recognition [4]

Research objective. The general trend toward deeper and more complex networks does not necessarily make them efficient in size and speed, yet applications such as robotics and augmented reality need timely recognition on limited platforms. The paper proposes small, low-latency models whose size can be matched to application constraints.

Proposed methodology. The core layer is the depthwise separable convolution, which factorizes a standard convolution into a depthwise convolution (one filter per input channel) and a 1 × 1 pointwise convolution. The authors derive a computation reduction of 1/N + 1/DK2, which for 3 × 3 kernels means 8 to 9 times less computation. The network has 28 layers, with batch normalization and ReLU after each convolution. Two global hyper-parameters shrink it further: a width multiplier α and a resolution multiplier ρ (input sizes 224, 192, 160 or 128). Because small models overfit less, training uses less regularization and augmentation than large models, and little or no weight decay on the depthwise filters.

Dataset used. ImageNet for the main comparisons; Stanford Dogs (fine-grained recognition, with pretraining on a larger noisy web set), a PlaNet-style geolocation task, a multi-attribute face classifier trained by distillation, COCO detection and FaceNet distillation. No fashion data.

Models and techniques. Comparisons include a full-convolution MobileNet, a shallower MobileNet, GoogLeNet, VGG-16, SqueezeNet, AlexNet and Inception V3. Distillation is used for the face-attribute and FaceNet experiments.

Evaluation metrics and results. Metrics are top-1 accuracy, multiply-adds, parameters, mean average precision and mAP. Selected ImageNet results are tabulated in the technical evidence tables below. Depthwise separable convolutions cost about 1 point of ImageNet accuracy relative to full convolutions (70.6% versus 71.7%) while using far fewer multiply-adds and parameters. On Stanford Dogs, MobileNet-224 reaches 83.3% top-1 versus 84% for Inception V3 (569 versus 5,000 million multiply-adds). In the face-attribute experiment, a distilled MobileNet reaches 88.7% mean AP against a 86.9% baseline with 1,600 million multiply-adds and 7.5 million parameters.

Table 2. Selected ImageNet efficiency trade-offs reported by Howard et al. [4].

Configuration	Top-1 accuracy (%)	Multiply-adds (M)	Parameters (M)
MobileNet 224, α = 1	70.6	569	4.2
MobileNet 224, α = 0.75	68.4	325	2.6
MobileNet 224, α = 0.5	63.7	149	1.3
MobileNet 224, α = 0.25	50.6	41	0.5

Configuration	Top-1 accuracy (%)	Multiply-adds (M)	Parameters (M)
MobileNet α = 1, input 128	64.4	186	4.2
Full-convolution MobileNet	71.7	4,866	29.3

Strengths. The architecture is simple, and the two multipliers give a smooth, documented accuracy-versus-cost trade-off. The paper demonstrates use across classification, fine-grained recognition, attribute prediction and detection.

Limitations. The work is a preprint. Accuracy falls as the model or input shrinks (alpha = 0.25 reaches 50.6% on ImageNet). Efficiency is reported as multiply-adds and parameter counts; measured on-device latency is not tabulated in the sections reviewed. The smallest inputs tested are 128 × 128, and the fine-grained result relies on pretraining with a large noisy web dataset.

Relevance to SmartListing Copilot. (i) MobileNet is the efficiency-oriented candidate: the proposal evaluates end-to-end response time and ships the system as a FastAPI/Streamlit service, so a smaller backbone is a legitimate alternative to ResNet. (ii) The resolution-multiplier results show accuracy declining with smaller inputs on ImageNet; the proposal's low-resolution images make this a risk to monitor, though not one this paper quantifies for fashion. (iii) The paper's guidance to regularize lightly applies to large-data training; on a small dataset the project must still verify overfitting directly. (iv) Whether depthwise factorization retains enough accuracy for five fashion types and colors is an empirical question for the experiments.

3.5 Radford et al. (2021): CLIP Zero-Shot Image Classification [5]

Research objective. Vision systems are trained to predict a fixed set of categories, so any new concept requires new labeled data. The authors ask whether learning from raw text paired with images, at large scale, yields transferable representations that let natural language specify new classifiers without task-specific training.

Proposed methodology. CLIP trains an image encoder and a text encoder jointly, predicting which of the N × N pairings in a batch of N (image, text) pairs are real, using a symmetric cross-entropy loss over cosine similarities with a learned temperature. The image encoder is a modified ResNet or a Vision Transformer; the text encoder is a Transformer (63M parameters, 12 layers, 512 wide, maximum length 76). Models are trained from scratch for 32 epochs with a batch size of 32,768, and the only augmentation is a random square crop. For zero-shot classification, the class names (embedded in a prompt such as "A photo of a {label}.") are encoded by the text encoder to form a linear classifier over image embeddings. Prompts can be specialized to the task and ensembled in embedding space.

Dataset used. Pretraining uses WIT, 400 million (image, text) pairs collected from public internet sources via 500,000 queries with up to 20,000 pairs per query. Evaluation covers more than 30 existing datasets, with a 27-dataset suite for linear-probe comparisons and the 12-dataset suite of [2]. The authors note that the pairs are unfiltered and uncurated, so the model learns social biases.

Models and techniques. Eight CLIP models (five ResNet-based, three ViT-based), with ViT-L/14 at 336 px as the best. Baselines include Visual N-Grams, a logistic regression on ResNet-50 features, BiT, SimCLRv2 and Noisy Student EfficientNet.

Evaluation metrics and results. Reported: zero-shot ImageNet accuracy rises from 11.5% (Visual N-Grams) to 76.2%, matching the original ResNet-50 without using its 1.28 million training labels (top-5: 95%). The default prompt alone improves ImageNet accuracy by 1.3%; an 80-prompt ensemble adds 3.5% more; across 36 datasets prompt engineering and ensembling add almost 5 points on average. Against a logistic regression on ResNet-50 features, zero-shot CLIP wins on 16 of 27 datasets (for example +28.9 on Stanford Cars) but loses by large margins on others (for example −37.1 on EuroSAT, −12.5 on Flowers102,

−11.3 on FGVC Aircraft). Zero-shot performance matches a 4-shot linear classifier on the same features on average, and a 16-shot classifier on ImageNet. For most datasets it remains 10 to 25 points below a fully supervised linear classifier on CLIP features (correlation 0.82). A linear probe on CLIP features beats the best public ImageNet model on 21 of 27 datasets, though the EfficientNet is slightly better on the low-resolution CIFAR-10 and CIFAR-100, which the authors suspect reflects CLIP's lack of scale augmentation. In a stress test on 515 low-resolution CCTV images, top-1 accuracy was 91.8% with distinct classes but fell to 51.1% when a close alternative caption was added (for example "parking lot with white car" versus "red car"); the close answer was chosen 40.7% of the time.

Strengths. Classifiers are defined by text with no task-specific training, results scale smoothly with compute, and zero-shot models show markedly smaller gaps under natural distribution shift (up to 75% smaller gaps on ImageNet variants). The linear-probe protocol gives a cheap, comparable adaptation path.

Limitations (acknowledged by the authors). Zero-shot CLIP is weak on fine-grained tasks (car models, flower species, aircraft variants), on abstract tasks such as counting, and near random on novel tasks absent from pretraining. It generalizes poorly to truly out-of-distribution data (88% on handwritten MNIST digits), it can only choose among the concepts supplied in the classifier, class design strongly affects behavior, and tuning on validation sets means development is not strictly zero-shot.

Relevance to SmartListing Copilot. (i) The results support CLIP's proposed role: a no-training baseline against which the supervised model is judged, and a fallback when the main model underperforms. (ii) The fine-grained weaknesses and the close-alternative stress test are directly relevant to similar shoe types and near color shades, so CLIP is expected to be weakest where the proposal expects ambiguity. (iii) The emphasis on class design and prompt wording applies to the "other product" prompt that the proposal uses for unsupported-category detection; that prompt's wording is a design decision to be selected on validation data only. (iv) The reported zero-shot-versus-linear-probe gap is the reason the optional linear probe is worth running once baselines are recorded. (v) CLIP is a parallel path from the image; nothing in [5] makes it a preprocessing stage for ResNet or MobileNet.

Synthesis. CLIP learns joint image–text representations but its proposed zero-shot use here is strictly an alternative visual classifier. It does not replace the independent title/description claim extractor or prove that a semantic mismatch has been detected.

3.6 Ramé et al. (2022): CoRe and Color Regression [6]

Research objective. The authors argue that existing garment-color methods are classification-based and cannot go beyond a list of predefined color names. They treat color as a regression problem, predicting the exact RGB value, and note that color is hard to estimate because of lighting, background pixels and multicolor garments.

Proposed methodology. A two-stage network. Stage one is a color classifier (VGG16 features, softmax over 72 color names, cross-entropy loss). Stage two refines the prediction by a weighted spatial pooling over the pixels of an illumination-corrected image. The weights come from two attention maps: an object-attention (DeepLabv3 pretrained on ModaNet clothing segmentation, then fine-tuned) that selects garment pixels, and a color-name-attention that keeps only pixels close to the color predicted in stage one, with a softmax temperature that can be fixed, learned, or predicted from the image. Illumination is estimated by a VGG16 head and corrected with the von Kries method after global histogram stretching. For multicolor garments the model predicts up to three colors and applies non-maximum suppression to remove near-duplicates. The regression loss is the Euclidean distance in CIELAB space, summed with the classification loss.

Dataset used. A newly collected dataset of 30,269 garments (5,363 coats, 8,166 dresses, 3,991 pants, 6,871 shoes, 5,878 tops); 2,000 images each for validation and test. Each garment was labeled by a single operator with its exact RGB value, defined as the color the operator believed it would show in ideal white

light, not as it appears in the photo. Color names are derived by nearest neighbor in LAB space, and multicolor items are tagged in decreasing order of importance. The paper does not state that the dataset is publicly released.

Models and techniques. VGG16 backbone for the illumination and color-name heads; DeepLabv3 for object-attention. Baselines: (a) unsupervised K-means clustering on pixels after background removal; (b) "Colorname RGB", an extension of an earlier multi-label color-naming approach that predicts color names and maps them to RGB. Training: Adam, learning rate 0.0001, batch 16, 50 epochs, random cropping and translation, class weights for imbalance.

Evaluation metrics and results. The metric is the percentage of predictions whose CIEDE2000 distance to the annotated color is below a threshold (10, 20, 30, 40). Reported (Table 1 of [6]); Selected threshold results are tabulated in the technical evidence tables below. When all colors of a garment are scored, the same three rows reach 19 / 31 / 37 / 42, 34 / 52 / 64 / 68, and 54 / 68 / 71 / 73. Each component (object-attention, color-name-attention, illumination module, predicted temperature) improves the result in the authors' ablation.

Table 3. CoRe main-color threshold success rates (%) at CIEDE2000 distance cutoffs [6].

Method	ΔE ≤ 10	ΔE ≤ 20	ΔE ≤ 30	ΔE ≤ 40
K-means after background removal	47	72	83	89
Colorname RGB	51	72	87	92
CoRe, best configuration	73	90	93	95

Strengths. It isolates the three failure causes of naive color averaging (illumination, background or neighboring-garment pixels, multicolor garments) and addresses each with a separate component. It uses a perceptual color space and a perceptual distance (CIELAB, CIEDE2000). The ablation shows what each component adds.

Limitations. The paper has no explicit limitations section; the following is this review's assessment. It is a short workshop paper. Labels come from a single annotator. The dataset is not stated to be public, so the results cannot be reproduced on it. No run-to-run variance is reported. Input image resolution is not stated in the text reviewed. The task (continuous RGB regression) differs from catalogue color classification. The pipeline needs a segmentation network, an illumination estimator and a VGG16 backbone, which is heavy for a small project.

Relevance to SmartListing Copilot. (i) The three failure causes it names match the lighting, background and multi-color rows of Table 9, so they are now supported by a color-specific source rather than analysis alone. (ii) The K-means baseline reaches 47% of main colors within $\Delta E \leq 10$ even after background removal, against 73% for the best learned model; this supports using the dominant-color estimate as a cross-check, not as the predictor (proposed). (iii) Its use of CIELAB and CIEDE2000 supports computing the dominant-color check in a perceptual space, and ΔE distance is a candidate way to define near-shade tolerance; any threshold is proposed and must be fixed on validation data. (iv) Its first stage is a color-name classifier over many names, which is the same shape as our color head over baseColour or color families; the regression stage is not needed because the comparison engine works with color families. (v) Its segmentation-based object-attention is not directly applicable: the proposal lists no mask annotations, and segmentation on 80 × 60 images is an untested option. (vi) Its labels follow an "ideal white light" definition, whereas baseColour is a catalogue-assigned label, so the label-audit step remains necessary. (vii) The authors tie their input-dependent temperature to calibration, which loosely connects to the project's confidence-calibration plan.

3.7 Confidence calibration and the dataset evidence boundary

Guo et al. [7] review confidence calibration and report that temperature scaling can improve the correspondence between predicted probabilities and observed correctness on their classification benchmarks. The SmartListing proposal therefore calls for reliability diagrams and Expected Calibration Error (ECE), evaluated on held-out data rather than for a claim that softmax probabilities are inherently reliable. Calibration cannot repair a wrong product category or a mislabeled image, and a well-calibrated model can still make individual mistakes.

Fashion Product Images Small [8] is the only dataset approved for project training, validation, testing and pipeline evaluation. Its styles.csv links images by id and supplies articleType, baseColour and productDisplayName. Dataset properties not checked directly in EDA, including exact resolution and class balance, should remain provisional. DeepFashion, ImageNet, Rakuten, eBay and MEPAVE are research-source datasets only, not additional SmartListing training data.

4. NLP and Multimodal Attribute Research

Textual claims present a different error mode from visual predictions. Product titles are short, noisy and often syntactically incomplete; a token matching a color may describe another garment or accessory rather than the listed item. Multimodal research demonstrates that images can improve attribute extraction, but most systems merge the evidence to produce a single completed attribute, obscuring the disagreement that listing-quality review needs to expose.

4.1 Putthividhya and Hu (2011): Bootstrapped Product Attribute NER [9]

Research objective. Extract four attributes (brand, garment type/style, size, color) from short eBay clothing and shoe listing titles, where titles are capped at 55 characters, have little grammar, and contain typos and acronyms. The authors also want to grow the attribute dictionaries from unlabeled data and normalize extracted values.

Methodology. Attribute extraction is posed as token-level named entity recognition. Features include word identity, orthography, context words, character n-grams and dictionary membership. POS tags and punctuation were not used because titles are not complete sentences. In the second part, a seed dictionary of unambiguous brands auto-labels listings, a MaxEnt classifier learns the surrounding context patterns, and newly discovered values are added back to the dictionary. Extracted values are normalized by character n-gram similarity.

Dataset. 105,335 men's and 72,628 women's clothing listings from eBay (1,380,337 word tokens, about 7.76 words per title). A sample of 1,000 men's listings was labeled and manually verified (7,921 labeled tokens) with a 90/10 train-test split.

Models. HMM, MaxEnt, SVM (each with Viterbi smoothing) and CRF.

Evaluation and results. On the 9-class tag task (in the source paper), accuracy was 83.82% for HMM, 88.13% for MaxEnt with Viterbi, 89.47% for SVM with Viterbi and 93.35% for CRF. Bootstrapping found new brands with 90.33% precision among the top 300 predicted (women's) and 89.5% among the top 200 (men's, 179 correct); for garment types, 54 of the top 60 were correct. Adding 200 discovered brands to the seed list raised NER accuracy from 83.56% to 92.75% on a test set containing no original seed brands, and from 90.02% to 93.66% on a random test set. Brand variants such as misspellings accounted for over 20% of listings, and n-gram matching (best n = 3 or 4) normalized them well, while Jaro-Winkler distance gave over 50% incorrect matches.

Strengths. It documents concrete ambiguity cases that a dictionary approach must handle: “camel” is both a brand and a color, “blazer” is both a brand and a garment type, and “inc” is both a brand and an abbreviation for inches. It also shows that misspellings and seller-specific abbreviations (BNWT, jkt, pjs) are common.

Limitations. The data are short English eBay titles from 2011, pre-neural. The labeled set is small (1,000 listings). The reported accuracy is over token tags, so it does not by itself give entity-level precision and recall, and the majority “not an attribute” class can inflate it. Discovery precision was checked manually on top-ranked outputs only. The authors note that with small training sets the classifier over-weights dictionary membership and misses new values. There is no image, no description text, and no non-English language.

Relevance to SmartListing Copilot. This is the closest published precedent for a dictionary-and-rules claim extractor over short fashion titles. It supports three design points: keep one controlled vocabulary per attribute with aliases mapped to a canonical value; resolve ambiguous words by context instead of by lookup alone (for example, “camel” as a color versus a brand); and expect misspellings and abbreviations, which fuzzy matching can normalize but also mis-normalize. The MVP does not train an NER model, so the CRF and bootstrapping results are evidence for what a later extension could achieve, not for the MVP itself.

Integrated interpretation. This publication demonstrates that dictionaries can support supervised NER, not that a standalone deterministic extractor has achieved the quoted CRF accuracy. The project should therefore test its rules as rules, report claim-level precision/recall and record the matched span and normalization rule rather than transferring token-tagging scores.

4.2 Zhu et al. (2020): M-JAVE and Gated Multimodal Extraction [10]

Research objective. Product attribute values are incomplete on e-commerce platforms; the authors report fewer than 8 attributes per clothing product on average against more than 40 possible ones. They aim to predict which attributes a description mentions and extract their values, using the product image to help where text is ambiguous.

Methodology. M-JAVE encodes the description with BERT and the image with a pretrained ResNet-101 (one global vector plus a 7x7 grid of 49 regional vectors). A global-gated cross-modality attention layer lets text tokens draw on image information, and a learned gate suppresses image signals that are irrelevant to a word. Attribute prediction is a multi-label classification. Value extraction is BIO sequence labeling, conditioned on the predicted attributes and on regional-gated image attention. Both tasks train jointly, with a KL-divergence term that penalizes disagreement between the attribute and value outputs.

Dataset. The authors annotated MEPAVE, a Chinese e-commerce set from JD.com: 87,194 text-image instances from 34,041 products in seven categories (clothes, pants, dresses, shoes, boots, luggage, bags), 26 attribute types and 2,129 values. Annotators also checked each text-image pair and removed unqualified ones. Agreement on 1,000 triple-annotated instances was 92.83%. Split: 71,194 train, 8,000 validation, 8,000 test. The English MAE dataset was used for an extra check.

Models and baselines. Word-matching (WSM), Sep-BERT, RNN-LSTM, Attn-BiRNN, Slot-Gated, Joint-BERT and ScalingUp, plus the text-only variant JAVE.

Evaluation and results. F1 on attributes / values (in the source paper): M-JAVE (BERT) 90.69 / 87.17; text-only JAVE (BERT) 87.98 / 84.78; Joint-BERT 86.93 / 83.73. Table 6 shows that image input must be gated: removing the global visual gate drops value-extraction F1 to 80.32, below the text-only model’s 84.78. In an adversarial test that swaps in a random incongruent image, F1 falls significantly (Fisher’s method, $p < 0.0001$ for both tasks), which the authors use to show the model really uses the image. Moving from formal product text to oral customer-service dialogue, M-JAVE outperformed JAVE by 4.31 (attribute) and 5.70 (value) F1 points. On the English MAE subset whose values appear in the text,

accuracy was 75.01% against 72.96% for the earlier model; on values that appear only in the image it was 52.11%, which the authors call risky for practical use. Per-attribute plots (their Figure 4) show Color among the appearance-related attributes handled well.

Strengths. Public code and data; ablations for every component; the paper explicitly shows both the benefit and the harm of naive visual fusion; and the incongruent-image protocol is a reusable evaluation idea.

Limitations. Chinese-language data from one platform. Unqualified text-image pairs were removed from the dataset, so the model is never trained or tested on contradictory pairs. The model needs 87k annotated instances. The task is to complete attributes, not to decide whether the listing is wrong.

Relevance to SmartListing Copilot. Two points matter directly. First, the paper's own example shows why claim extraction must be attribute-aware: in "golden band collar shirt paired with black shoes," the word "black" describes another item, not the product color. A rule-based extractor will meet the same trap and needs a documented rule for colors that modify other nouns. Second, the incongruent-image test is a published precedent for evaluating whether a system really depends on the image, and it parallels SmartListing's planned synthetic wrong-color and wrong-type listings. It evaluates image awareness, not contradiction detection, so it supports the test design but not any expected score.

Integrated interpretation. An incongruent-image perturbation tests whether M-JAVE uses the visual modality; it does not show that M-JAVE detects a seller's false statement. This difference between image dependence and contradiction classification is central to the proposed evaluation.

4.3 De la Comble et al. (2022): Rakuten Modality Attention [11]

Research objective. On Rakuten-Ichiba, most sellers give only free text, so color and material attributes are often missing. The paper asks which model, which modality and which fusion method work best for predicting these attributes from the product text and image.

Methodology. Text and image are first modeled separately. The text model is BERT (a Japanese checkpoint for Rakuten data) with a plain linear classifier over the summed token vectors; the authors found that deeper classifiers hurt and that BERT needs gentle fine-tuning with a warm-up and cosine learning-rate schedule. The image model is a pretrained DenseNet-121 with a linear classifier. The two are combined by a late-fusion "modality-attention merger": a small network assigns each product a text weight and an image weight (summing to 1), which scale the two representations before classification. Because the merger can ignore one modality entirely ("modality collapse"), a KL-divergence term pushes the weights toward a uniform split.

Dataset. Rakuten-Color (6,460,720 products, 19 color classes, multi-label) and Rakuten-Material (6,725,281 products, 57 classes), both drawn from five fashion-related categories; a public Rakuten ENS color challenge set (250,000 products, 19 classes); and two non-fashion benchmarks (MM-IMDB, UPMC Food-101). Labels are the values some merchants entered; only the first image and the concatenated title and description are used.

Evaluation metrics. F1 and recall at 95% precision (R@P95), chosen because wrong attribute values can make products unreachable, so the goal is maximum coverage at high precision.

Results. Table 2 (F1 / R@P95, in %): on Rakuten-Color, image-only 61.8 / 8.0, text-only 79.3 / 57.3, multimodal 84.5 / 66.0. On Rakuten-Material: 55.5 / 5.0, 84.6 / 60.4, 87.0 / 68.0. On Rakuten-ENS: 61.5 / 13.5, 69.2 / 35.0, 76.6 / 43.2. Table 3 shows simple concatenation of the two modalities scored 78.6 / 52.6 on Color, below the text-only model, while the modality-attention merger reached 84.5 / 66.0. Modality collapse appeared on Material without the KL term and was reduced with it. A human check of 5,000 sampled products judged the model's predicted attributes correct for 98%, which supported deployment on Rakuten-Ichiba.

Strengths. Very large real data; a systematic comparison of text-only, image-only and fused models; per-product modality weights that can be shown to a human (for example, 85% text weight when "Navy" is in the title, 93% image weight when the title says nothing about color).

Limitations. Ground truth is merchant-entered and therefore noisy. Image-only color prediction was weak even at this scale; the authors attribute this to images showing several color variants and carrying overlaid text. The task is to fill in missing attributes, not to detect a wrong one. Japanese text, 6.4 million training products and a yes/no human check that is not an independent label set all limit transfer.

Relevance to SmartListing Copilot. The paper supports three decisions. (1) Text and image each win on different products, so keeping them as separate evidence streams is justified. (2) When the text names no color, the image is the only evidence; SmartListing treats this as a missing-attribute case, not a mismatch. (3) A precision-targeted operating point such as R@P95 is a sensible way to choose thresholds, provided the target is set on validation data and frozen before testing. Its weak image-only color results also reinforce the CV evidence that color from low-resolution catalog images needs an "ambiguous / please verify" outcome.

Integrated interpretation. The improvement from 79.3 to 84.5 F1 on Rakuten-Color [11] supports complementary information in the two streams under that dataset's labels. A fused output that down-weights conflicting evidence is not, by itself, an explainable listing inconsistency alert.

4.4 Khandelwal et al. (2023): Generative MXT [12]

Research objective. Sellers often leave attributes blank or fill them wrongly. The authors want one scalable model that handles thousands of (product type, attribute) pairs, can output attribute values never seen in training (zero-shot), and can infer values that the text never mentions (value-absent), using distant supervision from the catalog instead of human labels.

Methodology. Attribute extraction is posed as question answering: the question is the attribute name (for example, "color"), the context is the product type plus the text, and a generated answer is the attribute value. MXT has three parts: a T5 text encoder-decoder; a Multimodal Adaptation Gate (MAG) that shifts the text embeddings using ResNet-152 image features; and an Xception-based module whose attribute-specific visual embeddings are fused through multi-head cross-attention. The T5 decoder generates the value.

Dataset. The 30PT set (30 product types, 38 unique attributes; 569k training and 84k validation products; a 73k-product test set from two product types) and the public E-commerce5PT benchmark (22 product-attribute pairs over 5 product types; 273,345 training and 4,259 test products). The text is English.

Models and baselines. CMA-CLIP (a CLIP-based multimodal classifier) and NER-MQMRC (a BERT reading-comprehension extractor), plus ablations without MAG, without Xception, and with single-product-type training.

Evaluation and results. The metric is recall at 90% precision (R@P90), plus F1 on E-commerce5PT. The abstract reports absolute R@P90 gains of 10.16 and 6.9 points over prior state of the art. Table 1 shows the multi-product-type MXT beating NER-MQMRC on the top-5, top-10 and top-15 attributes for both product types (for example +22.33% for the top-5 attributes of one type), but on E-commerce5PT, where every value is written in the text, its F1 gain is mixed: +15.56 and +13.23 for two product types, and -1.47 and -7.89 for two others. It outperformed NER-MQMRC on 16 of 22 attributes. Zero-shot: of 970 test pairs unseen in training, 124 were generated correctly. Value-absent inference is illustrated by generating "mini" for item length from the image. The authors report deployment for more than 10,000 product-attribute pairs and more than 150 million extracted values.

Strengths. One model for many attributes; handles values absent from the text; an honest ablation table; and a candid description of deployment problems, such as unnormalized catalog values and the cost of annotating thresholds.

Limitations. Stated by the authors: English only; the pretrained T5 tokenizer splits fashion terms such as "skater midi dress" into meaningless pieces; catalog values are noisy and need heuristic merging; evaluation relies on catalog data instead of human labels; numeric attributes suit extractive models better. In addition, the paper reports no color-specific score, and generated values can come from the image alone, a setting in which [10] reported much lower accuracy (52.11%, for a different model on a different dataset) than for values stated in the text.

Relevance to SmartListing Copilot. MXT is a contrast case. It is generative and trained at industrial scale, which is outside the MVP: the scope excludes LLM-generated content, and a generated value cannot be traced to a phrase in the listing. Two ideas still transfer. The value-absent notion maps to SmartListing's Missing Attribute Detection, where the image supplies a bounded suggestion that fills a fixed template instead of free text. The paper's deployment notes also show that value normalization and precision thresholds are practical problems, which supports SmartListing's color-family mapping and threshold freezing.

Integrated interpretation. MXT is evidence about large-scale generative extraction, not a proposed SmartListing component. Its value-absent setting motivates bounded missing-attribute suggestions, but generation, material recognition and unconstrained attribute expansion remain outside the approved MVP.

4.5 Ribeiro et al. (2016): LIME and the Limits of Explanation [13]

Research objective. Explain individual predictions of any classifier so that people can decide whether to trust a prediction and whether to trust the model before deploying it.

Methodology. LIME (Local Interpretable Model-agnostic Explanations) perturbs the input around the instance being explained, queries the black-box model on the perturbed samples, weights them by proximity, and fits a sparse linear model on an interpretable representation (word presence for text, super-pixels for images) with at most K features. SP-LIME then selects a small, non-redundant set of representative explanations by greedy submodular optimization, which gives a global view of the model.

Dataset and models. Book and DVD review sentiment sets (2,000 instances each; 1,600 train, 400 test) with decision trees, logistic regression, nearest neighbors, SVM and random forests; the 20 newsgroups Christianity-versus-Atheism subset with an RBF-kernel SVM; a self-built religion test set of 819 web pages per class; Google's Inception network on images; and a deliberately biased wolf-versus-husky logistic regression (20 training images).

Evaluation and results. Faithfulness: LIME recovered more than 90% of the truly important features of two interpretable classifiers on both review sets (for example 92.1% for sparse logistic regression and 97.0% for a decision tree on books). Trust in predictions (in the source paper): average F1 of 91.8 to 96.7 across models and datasets, far above a greedy baseline (45.0 to 58.1). On 20 newsgroups, an SVM with 94.0% held-out accuracy scored only 57.3% on the independent religion set, because it relied on irrelevant tokens such as headers; a manually cleaned classifier scored 88.6% and 69.0%. Human subjects choosing between the two classifiers did so more accurately with explanations selected by SP-LIME than with random selection. In the wolf-versus-husky test, the share of 27 graduate students who trusted the bad model fell from 10 to 3 after seeing explanations, and those naming snow as a possible cue rose from 12 to 25.

Strengths. Model-agnostic; evaluated with both simulated and human users; and it shows concretely that held-out accuracy alone can hide a model that learned the wrong cue.

Limitations. Stated by the authors: the interpretable representation may be too weak to explain some behaviors, for example a model that predicts “retro” from sepia tone. A sparse linear surrogate may be unfaithful where the model is strongly non-linear even locally. An image explanation with Inception took about 10 minutes, and the selection step for images was not addressed.

Relevance to SmartListing Copilot. SmartListing’s explanations are built into the design: every warning carries a reason code, the claimed value, the predicted value, a confidence and a severity, so rule-derived explanations are faithful by construction. LIME is a different tool, a post-hoc explainer for opaque components such as the CV classifier or the lightweight mismatch classifier. It is not part of the MVP and is cited as background and as an optional extension. Its more useful lesson for the project is methodological: the 94.0% versus 57.3% result is a reminder to keep synthetic and manually built test results separate and to inspect disagreements, which is already a success criterion of the project plan.

Integrated interpretation. A rule trace that says which title span and visual prediction conflicted is different from a post-hoc surrogate explanation of a black-box classifier. LIME can inform later CV auditing but is not a mandatory part of the current implementation.

5. Comparative Scientific Synthesis

5.1 Comparison across recognition, extraction and completion tasks

The visual and language studies should not be ranked by a single percentage. DeepFashion’s top-k categories, ResNet’s ImageNet error, MobileNet’s accuracy–compute curve, a token-level NER tagger, and multimodal product-value extractors measure different targets. More fundamentally, the best completion model may eliminate conflict by deciding which modality to trust, while the best consistency checker must preserve that conflict long enough to report it.

Table 4. Cross-domain evidence matrix; all findings belong to the original publications.

Evidence cluster	Principal studies	What is established	What remains unproven for SmartListing
Fashion recognition	DeepFashion [1]; CoRe [6]	Clothing category supervision and garment-pixel-aware color estimation can be useful; color is sensitive to scene conditions.	Reliable five-type and catalog-color recognition on the small low-resolution dataset.
Backbone transfer	Kornblith [2]; ResNet [3]; MobileNet [4]	Pretrained visual features transfer in many settings; depth and compute impose trade-offs.	Best backbone, freeze/fine-tune strategy and native-resolution failure rates.
Visual–language prior	CLIP [5]	Text-defined image categories allow zero-shot baselines; prompt choice matters.	Reliable distinction between near shades and neighboring fashion types.
Short-text claims	Putthividhya and Hu [9]	Ambiguity, aliases and misspellings materially affect attribute tagging.	Quality of a pure rule-based bilingual claim extractor without training.
Multimodal value extraction	M-JAVE [10]; Rakuten [11]; MXT [12]	Gated/weighted fusion can improve product-value completion on original datasets.	Detection and explanation of deliberate image–text contradiction.
Explainability	Calibration [7]; LIME [13]	Confidence needs calibration; local surrogates can reveal misleading cues.	Seller understanding and correctness of SmartListing reason-code warnings.

To preserve research-specific comparisons without confusing the original experiments, Tables 5 and 6 summarize the principal Computer Vision and NLP/multimodal papers separately. The values are selected examples from the cited studies, not scores obtained by SmartListing Copilot; detailed methods, caveats, and additional findings remain in Sections 3 and 4.

Table 5. Focused comparison of the six Computer Vision studies and their original evidence.

Publication / task	Data and approach	Published evidence / metric	Limitation and SmartListing implication
DeepFashion — Liu et al. (2016) [1]	800,000+ clothing images; 50 categories; FashionNet with VGG-16, landmarks and attribute supervision.	FashionNet category top-3 / top-5 accuracy 82.58% / 90.17%; benchmark-specific top-k results.	No dedicated color group; rich landmark labels and large-domain imagery do not transfer directly to styles.csv. Supports label auditing and product-ID split.
ImageNet transfer — Kornblith et al. (2019) [2]	Image classification transfer studies; frozen pretrained features versus end-to-end fine-tuning.	ImageNet/transfer accuracy correlations: r = 0.99 (frozen) and r = 0.96 (fine-tuned), across sampled models.	Verified at abstract level in original CV review; correlation does not establish native-resolution fashion performance. Compare transfer regimes experimentally.
Residual networks — He et al. (2016) [3]	Residual shortcuts; ImageNet/CIFAR-10 and detection evaluations; ResNet-18/34/50/101/152.	Published ImageNet error and GFLOPs depend on depth and evaluation protocol; specific measurements preserved in Table 1.	Large-image architecture, no project-specific type/color evidence. Compare smaller ResNet checkpoints without presuming fine-grained gains.
MobileNet — Howard et al. (2017) [4]	Depthwise separable convolution; width and input-resolution multipliers; original ImageNet tests.	Selected ImageNet accuracy, multiply-add counts and parameter trade-offs appear in Table 2.	Original architecture preprint; compute–accuracy trade-off is established on another task, not on the five-category low-resolution scope.
CLIP — Radford et al. (2021) [5]	Contrastively pretrained on 400 million image–text pairs; prompt-defined zero-shot classifiers.	Zero-shot ImageNet top-1 accuracy 76.2% in the reported protocol; depends on model/prompt.	Weakness on fine-grained labels and wording sensitivity. Parallel no-training baseline/fallback, not a text–image contradiction detector.
CoRe — Ramé et al. (2022) [6]	30,269 garment images; color regression, garment attention and CIEDE2000 thresholds.	Main-color predictions within $\Delta E \leq 10$: best approach 73% versus 47% for K-means, on its dataset.	Short CVPR workshop study, different labels/metric; supports background/illumination caution, not proven catalog baseColour performance.

Table 6. Focused comparison of the five NLP, multimodal and explainability studies.

Publication / task	Data and approach	Published evidence / metric	Limitation and SmartListing implication
Putthividhya & Hu (2011) [9]	eBay short fashion titles; dictionary features and bootstrapped NER; 1,000 labeled titles.	CRF token-label accuracy 93.35% over 9 tags; discovery precision reported for top-ranked candidates.	Token accuracy is not entity-level claim F1; ambiguity/misspellings motivate normalization and context rules, not a trained NER requirement.
M-JAVE — Zhu et al. (2020) [10]	MEPAVE: 87,194 Chinese product text–image pairs; BERT/ResNet-101 with gated joint extraction.	Attribute / value F1 = 90.69 / 87.17 versus text-only 87.98 / 84.78; ungated value F1 80.32.	Incongruent pairs excluded from curated data; image-swap ablation tests use of image, not detecting seller contradictions.
Rakuten merger — De la Comble et al. (2022) [11]	6.46M Rakuten-Color products; BERT/DenseNet-121 weighted late fusion; merchant-provided labels.	Color multimodal F1 / R@P95 = 84.5 / 66.0; image-only 61.8 / 8.0; text-only 79.3 / 57.3.	arXiv preprint; noisy catalog labels and completion setting. Separate modalities are valuable; missing text is not automatically mismatch.
MXT — Khandelwal et al. (2023) [12]	30PT and E-commerce5PT industrial benchmarks; T5 generation with visual adaptation and cross-attention.	R@P90 gains of 10.16 and 6.9 points in reported contexts; 124 / 970 unseen test values generated correctly.	Industrial generative approach outside MVP; results vary by type and use weak catalog labels. Template suggestions remain bounded.
LIME — Ribeiro et al. (2016) [13]	Model-agnostic local surrogate explanations across sentiment, text and image tasks.	Important-feature recovery: 92.1% (sparse logistic regression) and 97.0% (tree) in one book-review setting.	Explanation fidelity may fail on nonlinear decisions; post-hoc model interpretation is distinct from directly inspectable rule traces.

Comparative inference. The Computer Vision evidence motivates an explicit accuracy–efficiency–calibration study of independently predicted Product Type and Color. The NLP studies demonstrate why context-sensitive normalization is needed, while multimodal fusion papers primarily optimize attribute recovery rather than label contradiction. Consequently, mismatch precision, abstention behavior and explanation correctness require separate end-to-end experiments.

5.2 Why joint attribute fusion is not the proposed inconsistency detector

Zhu et al. [10] demonstrate that unselective visual injection can harm extraction: dropping a global gate lowered value F1 below the text-only system. De la Comble et al. [11] show that simple modality concatenation can underperform text, and that an attention merger must guard against modality collapse. Khandelwal et al. [12] prioritize producing a value even when the text omits it. These are meaningful achievements for catalogue completion; they are not evidence of a detector that identifies which text claim contradicts the picture.

A disagreement-preserving architecture is therefore a bounded and testable design response, not a novel multimodal learning invention. The proposed CV outputs and NLP claims are normalized into a common type/color vocabulary, compared attribute by attribute, and retained alongside the reason code. A missing claim produces MISSING_ATTRIBUTE rather than COLOR_MISMATCH, while a near shade within the agreed color family is not treated as a definite error. The comparator must abstain or request review when the visual or text evidence is weak.

Table 7. Rakuten modality ablation results [11]; values are F1 / recall at 95% precision (%) within each dataset.

Dataset	Image only	Text only	Multimodal
Rakuten-Color, 6,460,720 products	61.8 / 8.0	79.3 / 57.3	84.5 / 66.0
Rakuten-Material, 6,725,281 products	55.5 / 5.0	84.6 / 60.4	87.0 / 68.0
Rakuten-ENS, 250,000 products	61.5 / 13.5	69.2 / 35.0	76.6 / 43.2

5.3 Product color, ambiguity and evidence thresholds

Color combines three challenges highlighted across the reviews: pixel-level lighting/background ambiguity in CoRe [6], weak image-only high-precision recall for Rakuten [11], and lexical ambiguity in short titles [9]. A canonical color-family normalization layer is therefore essential before comparison. Perceptual color-distance or dominant-color analysis may be a supplementary cross-check, but the CoRe RGB-regression system is not adopted as a required MVP feature.

Table 8. Selected original numerical findings are not comparable across studies.

Study / dataset	Measure	Reported result	Interpretive caveat
DeepFashion [1]	FashionNet category top-3 / top-5	82.58% / 90.17%	50 clothing categories; not top-1 or Macro-F1
Transfer study [2]	ImageNet vs transfer correlation	r = 0.99 frozen; 0.96 fine-tuned	Correlation across networks, not fashion performance
CLIP [5]	Zero-shot ImageNet top-1	76.2%	Prompts and dataset-specific class semantics matter
CoRe [6]	Main color within $\Delta E \leq 10$	73% best method; 47% K-means	Continuous color metric on proprietary-study labels
Product NER [9]	9-class token tagging accuracy	93.35% CRF	May be inflated by non-attribute class
M-JAVE [10]	Attribute / value F1 on MEPAVE	90.69 / 87.17	Chinese curated image-text pairs
Rakuten-Color [11]	F1 / R@P95 multimodal	84.5 / 66.0	Merchant labels; value completion, not mismatch
MXT [12]	Zero-shot unseen attribute values	124 correct of 970 unseen pairs	Value generation, not contradiction accuracy
LIME [13]	Model trust-selection F1	91.8–96.7	Simulated/human trust task, not seller explanations

5.4 Evidence that must remain separate

Confidence calibration measures whether predicted probabilities behave like observed frequencies, not whether a textual claim is true [7]. NLP rule ambiguity concerns the provenance and reliability of a matched phrase rather than a calibrated classifier probability [9]. A high-severity mismatch is justified only when each relevant evidence stream supports a specific, non-near-equivalent value. Explanation claims must likewise be qualified: explicit reason codes faithfully report the comparator's rule, not the correctness of the CV feature attribution or of the warning itself [13].

6. Unified Research Gap and Proposed Contribution

The eleven selected principal studies establish capable recognition, transfer learning, text extraction, multimodal attribute completion and methods for model explanations. Within the reviewed set, however, no paper evaluates a seller-facing pipeline that (i) separately identifies Product Type and Color in an image and in listing claims, (ii) preserves conflicting values instead of fusing them away, (iii) distinguishes contradictions from omissions and ambiguity, and (iv) explains a proposed correction while requiring seller approval. This is a gap within the selected evidence, not a claim that no such system has ever been published.

Table 9. Unified research-gap matrix and testable response.

Evidence-supported limitation	Integration risk	Proposed and testable response
Visual benchmarks differ from low-resolution catalog images [1]–[6].	False visual claims drive false mismatch alarms.	Compare supervised backbones with CLIP; evaluate per-type and per-color; audit resolution, ECE and abstentions.
Color reflects lighting, garment/background mixing and noisy labels [6], [11].	Spurious cross-family color mismatches.	Canonical color families, image-quality checks, cautious thresholds and manual near-shade cases.
Short titles contain spelling variation and ambiguous descriptors [9], [10].	Wrong item or attribute extracted from text.	Rule-level provenance, context-aware aliases and claim-level EN/AR evaluation.
Fusion papers optimize extraction/completion, not disagreement labels [10]–[12].	Text–image contradiction gets concealed.	Independent image and text payloads, rule comparator, lightweight learned baseline and per-error-type assessment.
Evidence explanations may not guarantee correctness [7], [13].	Authoritative-looking but incorrect seller warnings.	Reason-code trace, calibrated confidence, cautious “please verify,” seller Accept/Edit/Ignore.
Industrial datasets and synthetic experiments differ from real sellers [10]–[12].	Overclaimed external validity.	Split by product id before injection; report synthetic and manual held-out evaluation separately.

The proposed contribution is the integration and empirical evaluation of established methods in a narrowly scoped, explainable listing-quality workflow—not the invention of ResNet, MobileNet, CLIP, NER, multimodal attention, calibration, transfer learning or explainable AI. No novel algorithmic performance or real-marketplace impact is claimed before experiments.

7. Unified SmartListing Copilot Conceptual Framework

Figure 1 summarizes the approved evidence flow. The arrow from each evidence branch to the comparator is deliberate: CLIP is parallel to, not sequentially composed with, the supervised CV candidate; textual claims are extracted independently; and the proposed rules preserve source evidence so that the seller can inspect it.

Figure 1. Unified conceptual architecture for independent visual/text evidence and seller-controlled review (proposed, not implemented).



Figure note. The CV and NLP branches retain independent evidence until comparison. Boxes describe proposed modules and decisions; no component-performance or deployment claim is implied.

7.1 Proposed rule-level decision semantics

Table 10. Examples of proposed decision semantics; thresholds and severity values are not empirical results.

Situation	Image and text relation	Suggested treatment
Type agreement	Both confidently indicate the same normalized type.	No type warning.
Type conflict	Text "sandals"; image "boots", reliable predictions.	TYPE_MISMATCH and reviewer-facing source evidence.
Color conflict	Text "red"; image "blue", distinct families.	COLOR_MISMATCH only if evidence passes validated thresholds.

Situation	Image and text relation	Suggested treatment
Near shade	Text “navy”; image “blue” in shared family.	Near-equivalence; do not force a contradiction.
Missing claim	Image supplies color but listing says none.	MISSING_ATTRIBUTE, bounded suggestion, not a mismatch.
Unreliable observation	Blur, low image confidence or lexical ambiguity.	IMAGE_BLUR / LOW_CONFIDENCE or “please verify”; do not assert correction.
Out-of-scope item	Image likely outside selected five types.	UNSUPPORTED_CATEGORY and human review.

7.2 Boundaries between reportable evidence and proposed implementation

The proposal permits a compact multi-head visual classifier, an optional CLIP linear probe, image-quality flags, a heuristic Listing Quality Score, and a learned lightweight mismatch classifier. None is proven necessary or accurate by the cited studies on the project data. Attributes outside Type and Color—material, size, price and return risk—remain excluded; full multimodal transformers, open-ended generated fixes, LoRA, marketplace integration and autonomous publishing are deferred.

8. Evaluation Implications and Outstanding Research Questions

A scientific assessment must use the dataset’s exact product IDs to split training, validation and test data before generating inconsistent titles or descriptions. Duplicate and near-duplicate images must be checked across splits. Canonical color families, CLIP prompts, thresholds and severity policies should be established on validation data and kept fixed when reporting held-out results. The manual EN/AR evaluation set should use test-split images and be reported separately from synthetic errors to avoid conflating constructed and naturalistic cases.

Table 11. Evaluation matrix derived from the integrated literature and authoritative project scope.

Subsystem	Proposed measures	Reason for separate reporting
Visual Product Type	Accuracy, Macro-F1, per-class confusion, native-resolution checks	Fine-grained errors and imbalance are hidden by overall accuracy.
Visual Product Color	Accuracy, Macro-F1, per-color results, ambiguity/near-shade checks	Background and lighting can make broad accuracy misleading.
Calibration and uncertainty	Reliability diagram, ECE, accuracy–coverage, review-abstention rate	High confidence may not imply correctness.
Rule-based claim extraction	Claim-level precision/recall/F1 for English and Arabic, alias and ambiguous-text cases	Token accuracy in published NER work cannot substitute for extracted-claim validity.
Mismatch comparator	Precision, recall, F1 and false-alarm rate per type/error; rules versus lightweight classifier	A classifier trained on synthetic errors may learn injection artifacts.
Missing attributes	Precision/recall for missing Type or Color claims	An absent claim is not a contradiction.
End-to-end reviewer workflow	Scenario-level correctness, reason-code trace, seller confirmation and latency	Template grounding does not guarantee model correctness; user comprehension is not yet proven.

Table 12. Rule-based claim extraction: original evidence and proposed evaluation controls.

Literature observation	Proposed NLP behavior	Recommended case-based test

Literature observation	Proposed NLP behavior	Recommended case-based test
Title abbreviations and typographic noise [9]	Canonical type/color lexicon, aliases; record exact matched span	Alias and misspelling precision/recall
Ambiguous words such as “camel” and “blazer” [9]	Use narrow context rules; no unconditional dictionary match	Adversarial ambiguous-noun cases
Colors can describe an outfit accessory [10]	Associate color with the item noun phrase	Accessory-color negatives, e.g. shirt paired with shoes
Fuzzy matching can normalize or introduce errors [9]	Bounded edit-distance matching and lower rule confidence	False-normalization rate and unsupported forms
Text silent about color [11]	Return missing claim, never fabricate contradiction	Missing-color and missing-type scenarios
No reviewed multilingual model applies to this lexicon	English primary; Arabic fixed vocabulary, unsupported forms abstain	English and Arabic claim-level F1 separately

The first unresolved question is empirical: how much visual accuracy survives in the dataset’s native low-resolution images, and whether the supposed gains of transfer learning survive independent test splits. The second is linguistic: whether a deterministic vocabulary can recover enough actual attributes from productDisplayName and simulated descriptions without attributing accessory colors to the listed product. The third concerns decision quality: whether adding a lightweight mismatch classifier improves precision or recall beyond its own generating rules, particularly on a manually constructed set. Finally, confidence and a reason code may aid review, but seller understanding and any effect on returns are not established by this literature.

9. Conclusion

Taken together, the Computer Vision literature supports investigating rich clothing representations, transfer learning, ResNet/MobileNet efficiency choices, parallel CLIP baselines and careful handling of garment color. NLP and multimodal studies provide concrete evidence about short-title ambiguity, gated cross-modal extraction, the value of missing-attribute completion and the risks of indiscriminate fusion. Calibration and explainability research warn that a plausible confidence or rationale does not prove a correct decision. These contributions are complementary but not interchangeable.

The resulting SmartListing Copilot design is a deliberate composition of independently testable visual and textual evidence streams, an explicit Type/Color discrepancy comparator, transparent reviewer-facing reasons and seller-controlled template edits. The central research gap is not the absence of strong recognition models; it is the lack of validated, explainable contradiction checking in the narrowly defined evaluation setting reviewed here. Planned experiments must resolve whether that composition works under noisy color labels, incomplete short text, low-resolution images and synthetic-data limitations.

All reported numerical results belong to prior researchers and should remain attached to their original studies. The project has not yet demonstrated classification performance, mismatch-detection accuracy, explanation usefulness or measurable commercial benefit. These remain testable research questions rather than completed contributions.

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